INCH-POUND
MIL-DTL-7793/13B
22 May 2017
SUPERSEDING
MIL-M-7793/13A
19 October 2012

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL, 5 VDC, PRINTED CIRCUIT MOUNT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Reactivated after 19 October 2012 and may be used for new and existing designs and acquisitions.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-7793.

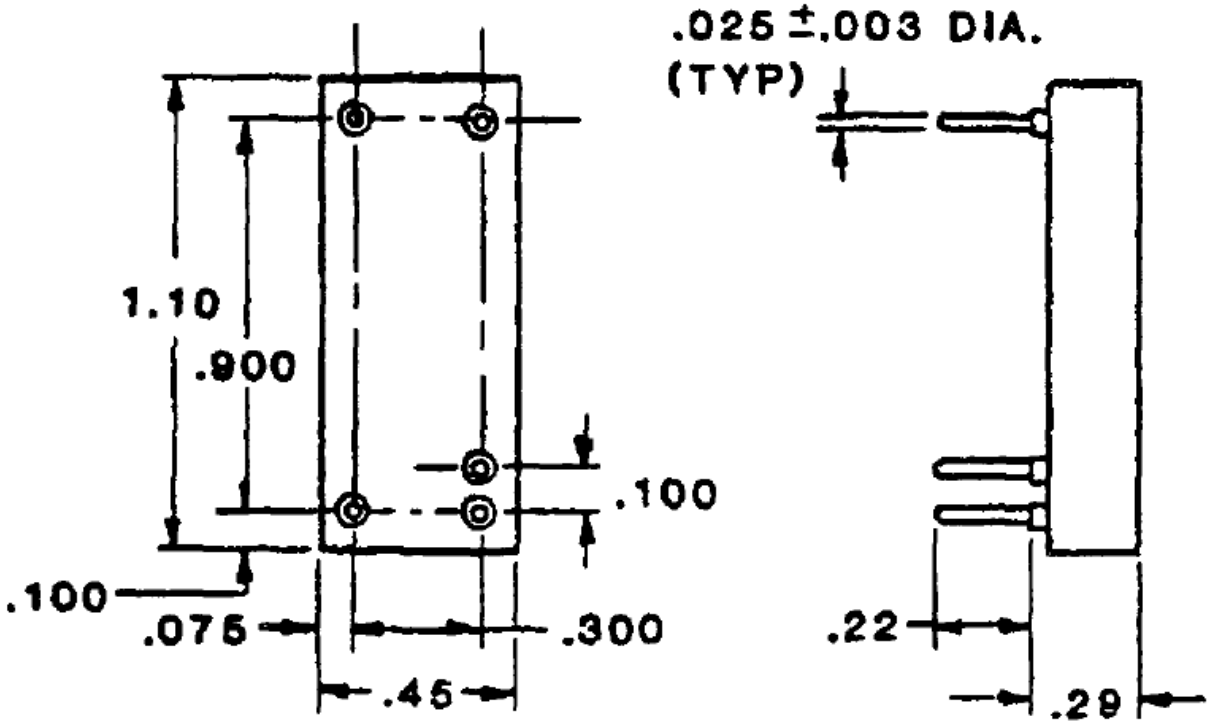


FIGURE 1. Bottom view.

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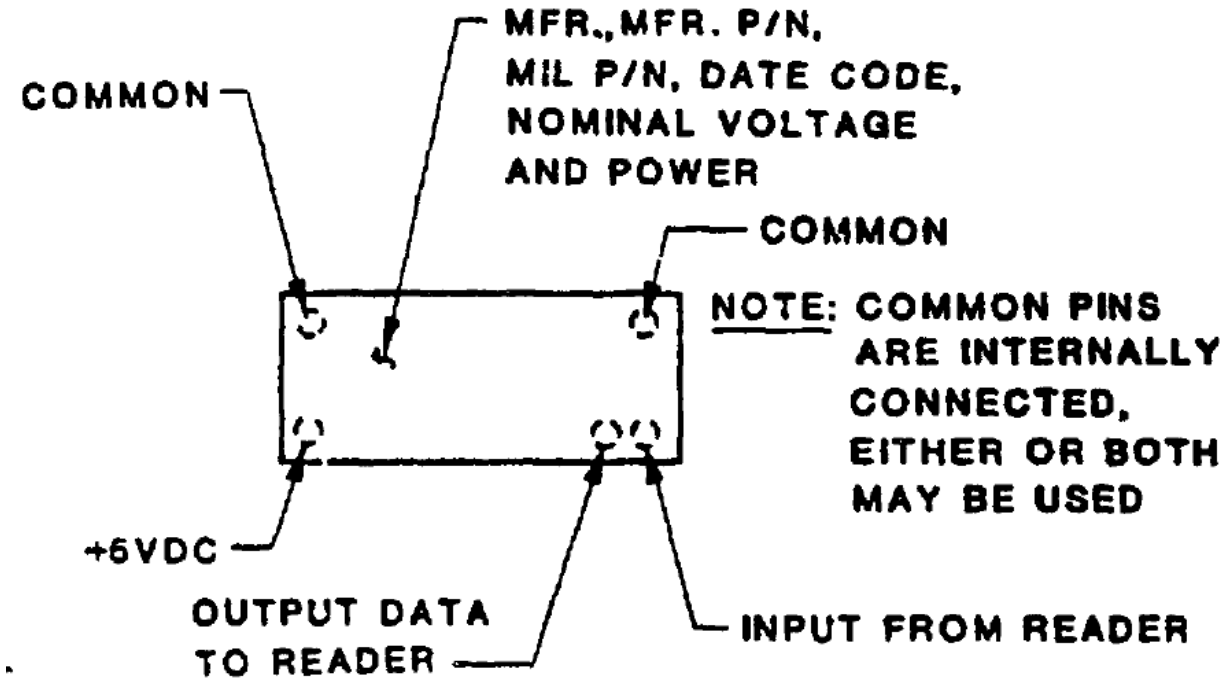


FIGURE 2. Top view and pin assignment.

REQUIREMENTS:

1. Dimensions. Dimensions are in inches. Unless otherwise specified, tolerances are ±.02 for two-place decimals and ±.015 for three-place decimals.

2. Dielectric strength. The meter shall be subjected to 600 VRMS (at 25º C ± 5º C) and 350 VRMS (altitude). The test voltage shall be applied between the power terminals (+5 VDC and common) and an external ground which contacts the meter case on the five sides without terminals.

3. Life accuracy. Meter accuracy shall be ±0.1 percent from -65 to +125° C and 4.5 to 10 VDC.

4. Power consumption. Power consumption at 5 volts direct current (VDC) shall not be greater than 2 milliwatts.

5. Ripple voltage. The meter shall operate normally when subjected to a 2-volt peak (4 volts peak-to-peak) ripple between 10 Hz and 10 KHz superimposed on 7.0 VDC.

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6. Operating temperature range. The operating temperature range shall be from -65° to +125° C.

7. Part number. The part number shall be M7793/13-1.

8. Time range. Total readout of the meters shall be 99999.99 hours.

9. Meter output. The output from the meter shall be in a serial binary coded decimal format. The output impedance shall be 100K ± 10 percent. A logical “zero” shall be between 0.0 and +0.2 volt. A logical “one” shall be between +3.3 and +6.6 volts. The output pin assignments are shown on figure 2 and the output data format is shown on figure 3.

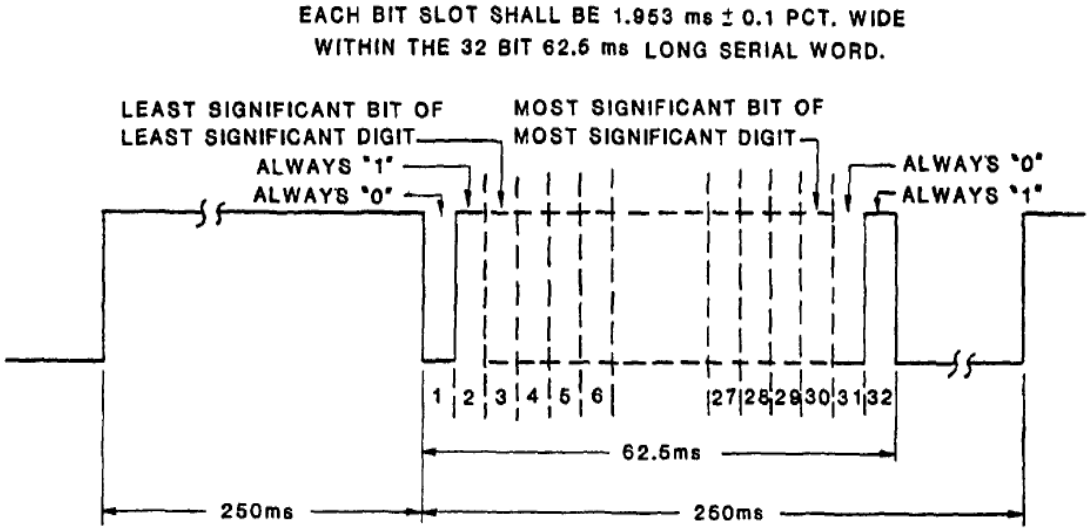


FIGURE 3. Output data format.

10. Transient protection. When operated on DC, no temporary or permanent degradation shall be produced in the meter when subjected to ±25 volt transients of 10 microseconds duration occurring at a 1 millisecond repetition rate.

11. Voltage range. The operating voltage shall be from 4.5 to 10 VDC.

12. Weight. Weight shall be not greater than 0.2 ounce.

13. Thermal shock. In lieu of thermal shock, 100 percent environmental thermal stress screening may be performed on 100 percent of meters supplied under this specification sheet.

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14. Environmental thermal stress screening test profile.

Temperature range: -65° C to +125° C
Temperature rate of change: 2° C/minute minimum (chamber air temperature)
Temperature dwell duration: 30 minutes minimum
Temperature cycles: 5 cycles minimum (one cycle consists of one dwell at each temperature extreme)
Power on/Equipment operating: The meter shall be turned on and off a minimum of 3 times at each temperature extreme (-65° C and +125° C) for each cycle. Power at 10 VDC

15. Qualification tests. Qualification tests shall be conducted in accordance with MIL-DTL-7793, except that the voltage, temperature, accuracy and thermal shock or environmental stress screening shall be as specified herein. In lieu of Thermal Shock, 100 percent Environmental Thermal Stress Screening may be performed on 100 percent of meters supplied under this specification sheet. Hermetic seal, salt spray and moisture resistance tests are not applicable.

16. Conformance inspection. Conformance inspection shall be conducted in accordance with MIL-DTL-7793, except that the voltage, temperature, thermal shock or environmental stress screening shall be as specified herein. In lieu of thermal shock, 100 percent environmental stress screening may be performed on 100 percent of meters supplied under this specification sheet. Hermetic seal, insulation resistance, and dielectric strength testing are not applicable.

17. Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

NOTES:

1. This unit is intended for use on printed circuit boards. All connections are expected to be made via printed wiring, and the output to be brought to a data collection point for system reading or to a single dedicated connector (see Note 2).

2. Elapsed time is read from the M7793/13-1 meter by mating the printed wiring board connection with the M7793/12-1 reader's connector and operating the reader.

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Navy - AS
Air Force – 71

Preparing activity:
Navy - AS
(Project 6645-2017-001)

Review activities:
Navy - OS, SH
DLA – GS 1

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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